



Making computers *talk*



Natural Language Processing lies at the intersection of linguistics and computer science, and promises to change the way we interact with computers. Forever.

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What is Natural Language Processing?

Natural language, or the language spoken by humans the world over, is a fascinating construct. Not only does each language have its own set of rules concerning grammar and semantics, but over time these languages spawn dialects that modify these rules in subtle ways. Computers, on the other hand are programmed using a set of instructions that are collectively called a programming language. These languages have their own 'grammar' (called syntax) and their own 'vocabulary' (called keywords) and bear little or no resemblance with natural language. While these programming languages operate on a very strict set of rules governed by logic, spoken language is often far more flexible, and any attempt to quantify these rules would fall under the domain of 'fuzzy logic'.

Natural Language Processing (NLP) is an attempt to use the concepts of linguistics and machine learning to allow computers to understand and communicate using natural languages. The earliest attempts at getting computers to understand human languages were in the 1950s, where machines were programmed to translate texts between different languages using a hard-coded set of rules. However, such attempts failed quickly owing to the imprecise and ever-changing nature of language. Recent

developments in the fields of machine learning using artificial neural networks have used statistical models that 'train' the software to understand the subtleties of language. Such systems can interpret natural languages more accurately, owing to their probabilistic approach, where they are trained using a set of known rules or words, and use this data to interpret new, raw data.

This form of NLP that uses machine learning models has proven far more successful than earlier approaches which modelled themselves after computer languages rather than human languages. The problem in this approach was that programming languages have fewer rules which are enforced more strictly than natural languages, which follow complex and ever changing rules of grammar. Language-specific semantics is another hurdle, as the literal meaning of words may not always be accurate.

Fans of The Office (US) will be familiar with Steve Carell's famous retort "That's what she said", which spawned a whole new genre of jokes. Inspired by this, two students at the University of Washington developed a system of identifying sentences which can be turned into a TWSS joke. Hit the link below to read the paper that they published, titled "That's What She Said: Double Entendre Identification". <http://www.aclweb.org/anthology-new/P/P11/P11-2016.pdf>

Machine learning models the problem from a human perspective. Humans tend to learn a new language faster using examples as opposed to strict rules, and NLP algorithms try to use this very fact to improve the accuracy of their systems.

Algorithms used in Natural Language Processing: The Markov Model and Parts of Speech

Most algorithms that classify text use complex learning algorithms that involve statistical models to extract meaning from text. An older, less accurate but considerably simpler technique used is the Markov model, which is used in classifiers such as Parts Of Speech (POS) tagging or sentiment analysis. POS is a system which tries to assign a tag to each word in a sentence, identify it as an adjective, verb, noun, article, singular, plural etc. Sentiment analysis is a technique that looks for certain words or 'identifiers' that determine the overall 'mood' or sentiment of the sentence. Such analysis is useful analyzing social media to gauge group sentiment.

One company putting sentiment analysis to good use on social media data is Paraktweet, which has launched two products. The first is called Bookvibe which gives book recommendations based on positive tweets about a tome, and the second is TrendFinder which helps entertainment companies gauge the audience's response to a movie by following the conversations around a movie or TV series. Parsing systems have also been developed